

In [1]:

```
pip install numpy
```

Requirement already satisfied: numpy in c:\users\hp\anaconda3\lib\site-packages (1.18.1)

Note: you may need to restart the kernel to use updated packages.

In [12]:

```
import numpy as np

array = np.array([2,4,6,8,10])  #List

print(array)

print(type(array))
print("Dimension of an array = " ,array.ndim)
```

```
[ 2  4  6  8 10]
<class 'numpy.ndarray'>
Dimension of an array =  1
```

In [3]:

```
print(np.__version__)
```

```
1.18.1
```

In [11]:

```
array = np.array((1,2,3,4,5))  #tuple
print(array)
print(type(array))
print("Dimension of an array = " ,array.ndim)
```

```
[1 2 3 4 5]
<class 'numpy.ndarray'>
Dimension of an array =  1
```

In [10]:

```
array = np.array((43))
print(array)
print(type(array))
print("Dimension of an array = " ,array.ndim)
```

```
43
<class 'numpy.ndarray'>
Dimension of an array =  0
```

In [13]:

```
array = np.array([ [4,5,6], [7,8,9], [1,2,3], [4,5,6] ])
print(array)
print(type(array))
print("Dimension of an array = " ,array.ndim)
```

```
[[4 5 6]
 [7 8 9]
 [1 2 3]
 [4 5 6]]
<class 'numpy.ndarray'>
Dimension of an array = 2
```

In [15]:

```
array = np.array([[ [4,5,6], [7,8,9], [1,2,3], [4,5,6] ] , [ [3,2,1], [9,8,7], [1,2,3], [4,5,6] ]])
print(array)
print(type(array))
print("Dimension of an array = " ,array.ndim)
```

```
[[[4 5 6]
 [7 8 9]
 [1 2 3]
 [4 5 6]]

 [[3 2 1]
 [9 8 7]
 [1 2 3]
 [4 5 6]]]
<class 'numpy.ndarray'>
Dimension of an array = 3
```

In [17]:

```
array = np.array((1,2,3,4,5), ndmin = 5)
print(array)
print(type(array))
print("Dimension of an array = " ,array.ndim)
```

```
[[[[[1 2 3 4 5]]]]]
<class 'numpy.ndarray'>
Dimension of an array = 5
```

In [22]:

```

array = np.array((1,2,3,4,5))
print(array[0])
print(array[3])

vareven = 0
varodd = 0
for element in array:
    if element%2 == 0:
        vareven = vareven + element
    else:
        varodd = varodd + element
print("Addition of even numbers:: ", vareven)
print("Addition of odd numbers:: ", varodd)

```

```

1
4
Addition of even numbers:: 6
Addition of odd numbers:: 9

```

In [24]:

```

array = np.array([ [4,5,6], [7,8,9], [1,2,3], [4,5,6] ])
print(array[0,0])

```

```

4

```

In [30]:

```

array = np.array([[ [4,5,6], [7,8,9], [1,2,34], [4,5,6] ] , [ [3,2,1], [9,8,7], [1,2,3], [4,5,6] ]])
print(array)
print(array[0,2,2])

```

```

[[[ 4  5  6]
  [ 7  8  9]
  [ 1  2 34]
  [ 4  5  6]]

```

```

  [[ 3  2  1]
   [ 9  8  7]
   [ 1  2  3]
   [ 4  5  6]]]

```

```

34

```

In [32]:

```

array = np.array([ [4,5,6], [7,8,9], [1,2,3], [4,5,65] ])
print(array[-1,-1])

```

```

65

```

In [40]:

```
array = np.array((1,2,3,4,5,4,3,2,1,4,5,32,22))
print(array[0:7])
print(array[0:])
print(array[0::2])
```

```
[1 2 3 4 5 4 3]
[ 1  2  3  4  5  4  3  2  1  4  5 32 22]
[ 1  3  5  3  1  5 22]
```

In [41]:

```
import numpy as np

arr = np.array([[1, 2, 3, 4, 5], [6, 7, 8, 9, 10]])
#slicing of 2-D array
print(arr[1, 1:4])
```

```
[7 8 9]
```

In [42]:

```
arr = np.array([[1, 2, 3, 4, 5], [6, 7, 8, 9, 10]])
print(arr[0:2, 2])
```

```
[3 8]
```

In [43]:

```
print(arr[0:2, 1:4])
```

```
[[2 3 4]
 [7 8 9]]
```

In [44]:

```
arr = np.array([[1, 2, 3, 4, 5], [6, 7, 8, 9, 10],[1, 2, 3, 4, 5], [6, 7, 8, 9, 10]])
print(arr)
```

```
[[ 1  2  3  4  5]
 [ 6  7  8  9 10]
 [ 1  2  3  4  5]
 [ 6  7  8  9 10]]
```

In [46]:

```
print(arr[0:4, 1:4])
```

```
[[2 3 4]
 [7 8 9]
 [2 3 4]
 [7 8 9]]
```

In [47]:

```
print(arr.shape)
```

```
(4, 5)
```

In [49]:

```
arr = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12])
twodarray = arr.reshape(3,4)
print(twodarray)
```

```
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
```

In [50]:

```
threedarray = arr.reshape(2,3,2)
print(threedarray)
```

```
[[[ 1  2]
   [ 3  4]
   [ 5  6]]

 [[ 7  8]
   [ 9 10]
   [11 12]]]
```

In [52]:

```
threedarray = arr.reshape(2,3,3)
print(threedarray)
```

```
-----
-
ValueError                                Traceback (most recent call last)
<ipython-input-52-02008f5ff555> in <module>
----> 1 threedarray = arr.reshape(2,3,3)
      2 print(threedarray)
```

ValueError: cannot reshape array of size 12 into shape (2,3,3)

In [53]:

```
threedarray = arr.reshape(3,2,2)
print(threedarray)
```

```
[[[ 1  2]
   [ 3  4]]

 [[ 5  6]
   [ 7  8]]

 [[ 9 10]
   [11 12]]]
```

In [55]:

```
array = np.array((1,2,3,4,5,4,3,2,1,4,5,32,22))
for var in array:
    print(var)
```

```
1
2
3
4
5
4
3
2
1
4
5
32
22
```

In [58]:

```
arr2d = np.array([[1, 2, 3, 4, 5], [6, 7, 8, 9, 10]])
for arr1d in arr2d:
    for element in arr1d:
        print(element, end = " ")
    print('')
```

```
1 2 3 4 5
6 7 8 9 10
```

In [61]:

```
arr3d = np.array([[ [4,5,6], [7,8,9], [1,2,3], [4,5,6] ],
                  [ [3,2,1], [9,8,7], [1,2,3], [4,5,6] ]])

for arr2d in arr3d:
    for arr1d in arr2d:
        for var in arr1d:
            print(var, end = " ")
        print('')
    print('')
```

```
4 5 6
7 8 9
1 2 3
4 5 6

3 2 1
9 8 7
1 2 3
4 5 6
```

In [68]:

```
import numpy as np
arr1d1 = np.array([34,55,67,88])
arr1d2 = np.array([54,88,37,34])

arr1d3 = np.concatenate((arr1d1, arr1d2))
print(arr1d3)
```

```
[34 55 67 88 54 88 37 34]
```

In [71]:

```
import numpy as np
arr2d1 = np.array([ [34,55],[67,88]])
arr2d2 = np.array([ [54,88],[37,34]])

arr2d3 = np.concatenate((arr2d1, arr2d2), axis = 0)
print(arr2d3)
```

```
[[34 55]
 [67 88]
 [54 88]
 [37 34]]
```

In [72]:

```
arr2d3 = np.concatenate((arr2d1, arr2d2), axis = 1) #Join two 2-D arrays along rows
print(arr2d3)
```

```
[[34 55 54 88]
 [67 88 37 34]]
```

In [77]:

```
#Sort
sortarr1d1 = np.sort(arr1d2)
print(arr1d2)
print(sortarr1d1)
```

```
[54 88 37 34]
[34 37 54 88]
```

In [83]:

```
arr = np.array(['rohan','sahil','joe','ajay'])
print(np.sort(arr))
```

```
['ajay' 'joe' 'rohan' 'sahil']
```

In [84]:

```
arr = np.array([True, False, True, False, True])
print(np.sort(arr))
```

```
[False False  True  True  True]
```

In [85]:

```
import numpy as np

arr = np.array([[3, 2, 4], [5, 0, 1]])

print(np.sort(arr))
```

```
[[2 3 4]
 [0 1 5]]
```

In [86]:

```
arr = np.array(['rohan', 'sahil', 'joe', 'ajay'])
arrboolean = ([True, False, True, False])

newarr = arr[arrboolean]
print(newarr)
```

```
['rohan' 'joe']
```

In [89]:

```
import numpy as np
array = np.array((1,2,3,4,5,4,3,2,9,4,27,33,22))

filteredarray = []

for var in array:
    if var%3 == 0:
        filteredarray.append(True)
    else:
        filteredarray.append(False)
newarr = array[filteredarray]
print(array)
print(filteredarray)
print(newarr)
```

```
[ 1  2  3  4  5  4  3  2  9  4 27 33 22]
[False, False, True, False, False, False, True, False, True, False, True,
 True, False]
[ 3  3  9 27 33]
```


In [90]:

```
import numpy as np
array = np.array((1,72,43,64,554,46,34,23,59,774,27,33,22))

filteredarray = []

for var in array:
    if var > 50:
        filteredarray.append(True)
    else:
        filteredarray.append(False)
newarr = array[filteredarray]
print(array)
print(filteredarray)
print(newarr)
```

```
[ 1  72  43  64 554  46  34  23  59 774  27  33  22]
[False,  True,  False,  True,  True,  False,  False,  False,  True,  True,  False,  F
alse,  False]
[ 72  64 554  59 774]
```

In []: